DRIFTING PACK ICE

Most of the sea ice is mobile pack ice that drifts on the polar oceans. In the Arctic Ocean the currents carry the ice across the North Pole, where the low temperatures make it grow thicker over many years. Eventually, it starts drifting away from the Pole, gets thinner, and finally melts into the ocean. This means that the marker on the ice indicating the North Pole is always moving with the ice, and has to be regularly relocated.

FROZEN VOYAGE

In the 1890s, Norwegian Fridtjof Nansen proved that ice drifted across the North Pole by allowing his specially strengthened ship *Fram* to become frozen into it. Over three years, the current carried the ice and ship across the top of the world, past the North Pole, until *Fram* broke free of the ice near Svalbard, Norway, in August 1896.

▸ SUPER-STRONG
An icebreaker can move through sea ice up to 6 ft (1.8 m) thick.

ICEBREAKER

All around the Arctic Ocean, frozen seas are cleared for shipping by powerful icebreakers. These specially strengthened ships have gigantic engines that drive them up over the floating ice, so their immense weight smashes the ice. Icebreakers also work around Antarctica, but less frequently because there are no major shipping routes there.